Shayan Shakeri

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PROFESSIONAL SUMMARY

AI Research Engineer at BMO's AI Centre of Excellence, building and owning LLM-agent systems: a graph-based agentic data layer built from the ground up, and the deterministic eval pipeline that caught a GenAI tool serving $200B+ AUM downplaying investment risk. Earlier, designed and validated hardware for an FDA Breakthrough-designated device now in multinational clinical trials. Co-founder of Merlyn Labs; open-sourced a flow-matching VLA integration for RLinf.

CORE COMPETENCIES

Agentic Systems | Agent Evaluation | LLM Fine-tuning | Reinforcement Learning | Python & C/C++ | PyTorch & JAX

WORK EXPERIENCE

BMO AI Centre of Excellence
Sep 2025 - Present

AI Research Engineer

Toronto, Canada

- Built a graph-based agentic system from the ground up to answer multi-hop relational queries across bank client data; since grown into a multi-agent data-layer harness.
- Uncovered systematic bias to downplay investment risk in a production GenAI tool serving $200B+ AUM wealth management.
- Built the deterministic agent eval pipeline that produced that finding: hundreds of synthesized inputs detecting misalignment invisible at anecdote scale.
- Building internal infrastructure that provisions LLM agents from a written role and scope definition.
- Designing evals for LLM agents that retrieve and reason over commercial banking and insurance policy.

Merlyn Labs
Aug 2025 - Present

AI Researcher & Co-Founder

Toronto, Canada

- Open-sourced a flow-matching VLA integration for RLinf, enabling RL training on the BEHAVIOR-1K suite in OmniGibson.
- Developing VLM judges that score rollouts into dense, context-dependent RL rewards that are difficult to game.
- Showed π0.5 generalization failures on LIBERO-PRO are recipe-induced overfitting; a conservative finetuning recipe doubled position-swap success (21% to 42%).

Epineuron Technologies
May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op

Mississauga, Canada

- Designed and assembled the PCBs and authored the IEC 60601-1 / ISO 13485 validation protocols for clinical test runs of PeriPulse, an FDA Breakthrough-designated device now in multinational clinical trials.
- Cut power draw through oscilloscope and power-analyzer testing, a 900% battery-life improvement.
- Modeled electromagnetic field penetration into peripheral nerves in COMSOL; the data set the electrode diameter.

PROJECTS

Stanford BEHAVIOR-1K Challenge
8th Place, Standard Track

Identified proprioceptive collapse as a critical VLA failure mode (60% masking improved task success up to 48%); found chunked execution outperformed temporal ensembling 3x. Trained on 10,000+ demonstrations covering 22 of 50 scored task types; published a technical report and a LessWrong analysis.

BardSong
Closed Alpha

Built the end-to-end pipeline: Groq speech-to-text, Gemini scene extraction, image generation, narrated video. Fine-tuned an LLM on D&D sourcebooks for narrative coherence; in closed alpha with 23 dungeon masters.

EDUCATION

M.Eng in AI & Robotics University of Toronto — Sep 2024 - Apr 2027 (anticipated)

B.Eng, Engineering Physics Co-op McMaster University — Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

Agents & ML: Agents, Agent Evaluation, Alignment Evaluation, LLM Fine-tuning, Reinforcement Learning, Imitation Learning, Flow Matching

Systems & Languages: Python, C/C++, PyTorch, JAX, OmniGibson, MuJoCo, ROS2, Sim-to-Real, PCB Design